

R Analysis Report

Data Modelling Project

September 26, 2026

Executive Summary

This report presents the results of the R-based analysis layer applied to the PostgreSQL warehouse core schema. The analysis covers customer tenure, revenue seasonality, fulfillment funnel, campaign lift, inventory health, pipeline health, and geographic splits.

1 Customer Tenure (01)

1.1 Overview

Analysis of SCD2 customer dimension revealing version count distribution and tenure spans.

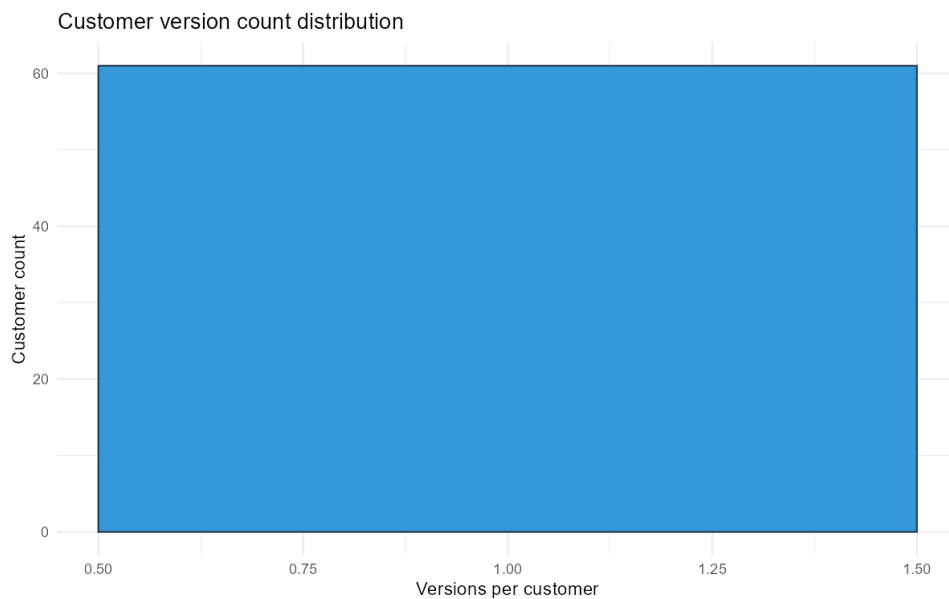


Figure 1: Customer Version Count Distribution

1.2 Key Findings

Distribution of customer versions across the SCD2 dimension.

2 Revenue Seasonality (02)

2.1 Overview

Monthly revenue patterns with STL decomposition for trend, seasonal, and residual components.

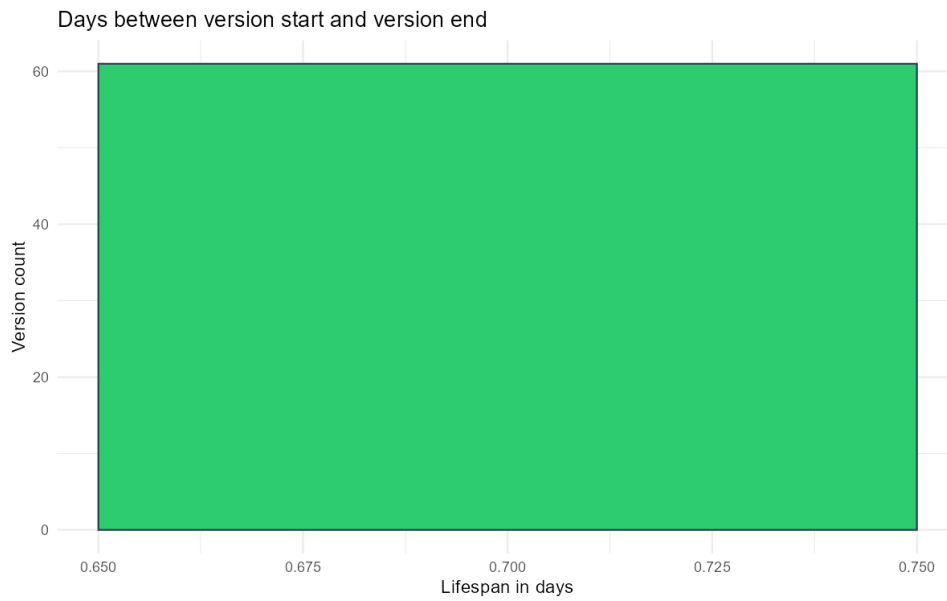


Figure 2: Version Lifespan Distribution in Days

2.2 Key Findings

Seasonal patterns in revenue data.

3 Fulfillment Funnel (03)

3.1 Overview

Order fulfillment conversion funnel with survival curve analysis for time-to-pay.

3.2 Key Findings

Bottlenecks in the fulfillment process.

4 Campaign Lift (04)

4.1 Overview

Spend versus revenue lift correlation across campaigns.

4.2 Key Findings

Correlation between campaign spend and lift.

5 Inventory Health (05)

5.1 Overview

Stock levels by category with ABC analysis.

5.2 Key Findings

Inventory health and stockout risk.

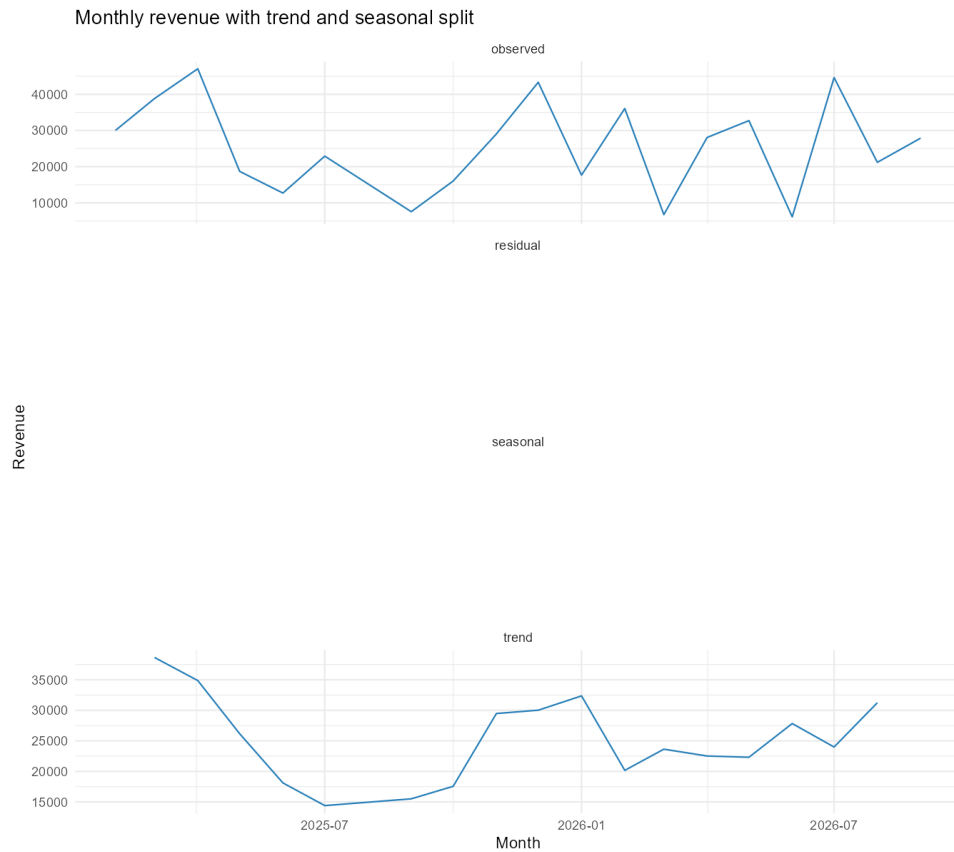


Figure 3: Revenue STL Decomposition

6 Pipeline Health (06)

6.1 Overview

Pipeline run duration trends with outlier detection.

6.2 Key Findings

Pipeline performance over time.

7 Geographic Split (07)

7.1 Overview

Order volume by state with ship-to vs bill-to divergence analysis.

7.2 Key Findings

Geographic distribution of orders.

8 Methodology

8.1 Data Source

All analysis queries read from the `core` schema in PostgreSQL via RPostgres.

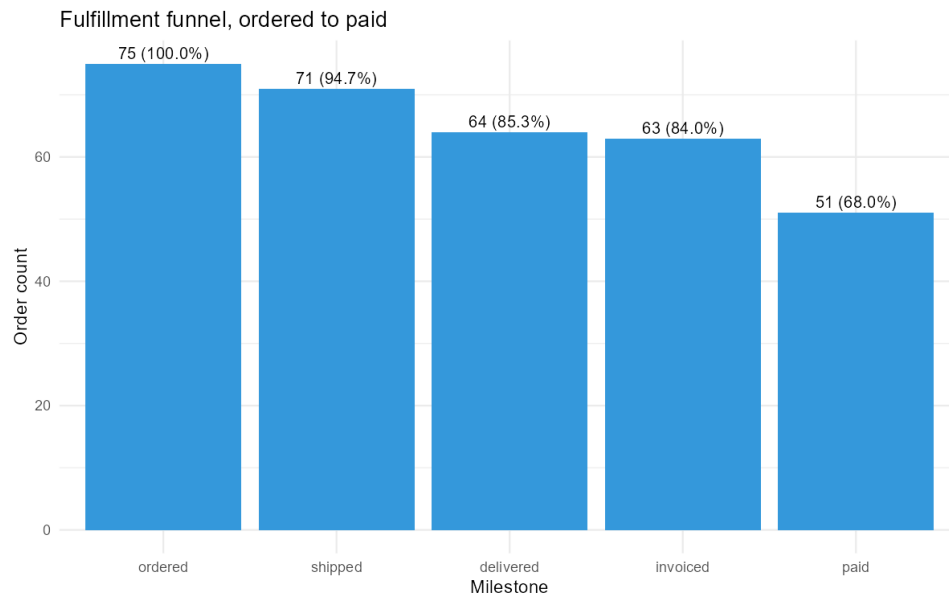


Figure 4: Fulfillment Funnel from Ordered to Paid

8.2 Analysis Approach

- EDA profiling via Quarto notebooks - Headless R scripts for batch analysis - Statistical analysis using tidyverse, survival, and forecast packages - Visualization via ggplot2 with consistent theme

8.3 Dependencies

R packages documented in `renv.lock`

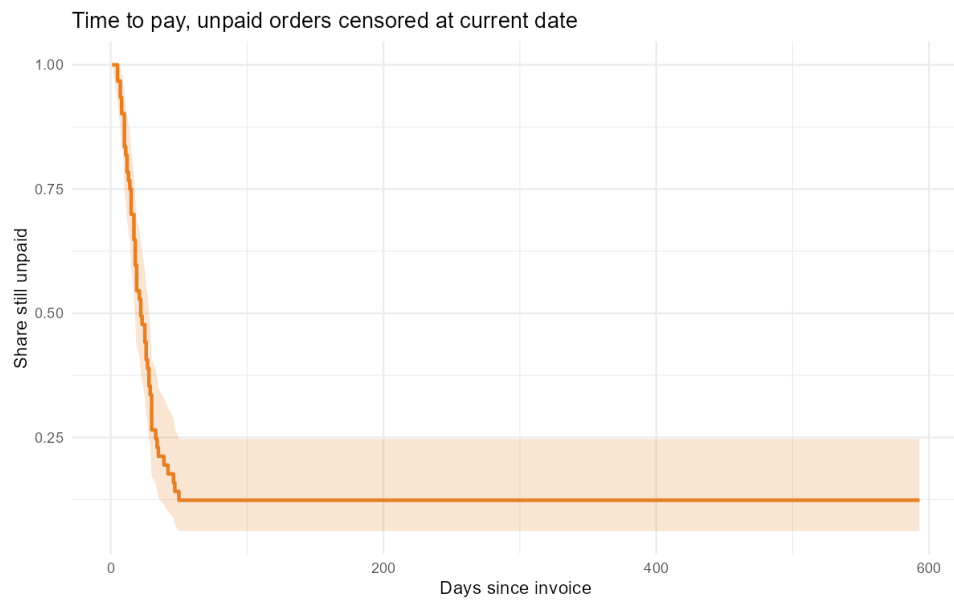


Figure 5: Time to Pay Survival Curve

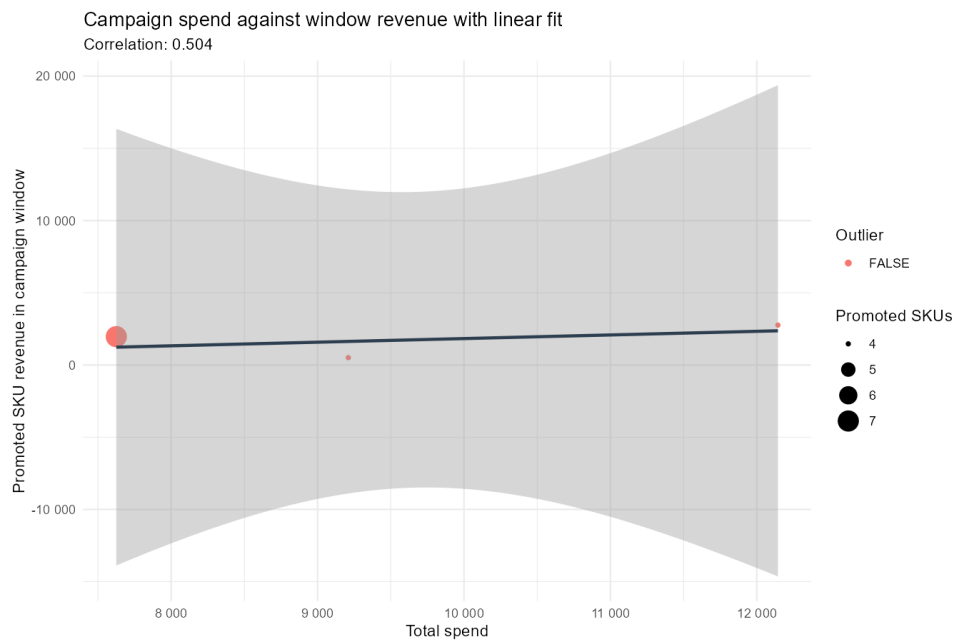


Figure 6: Campaign Spend Against Window Revenue

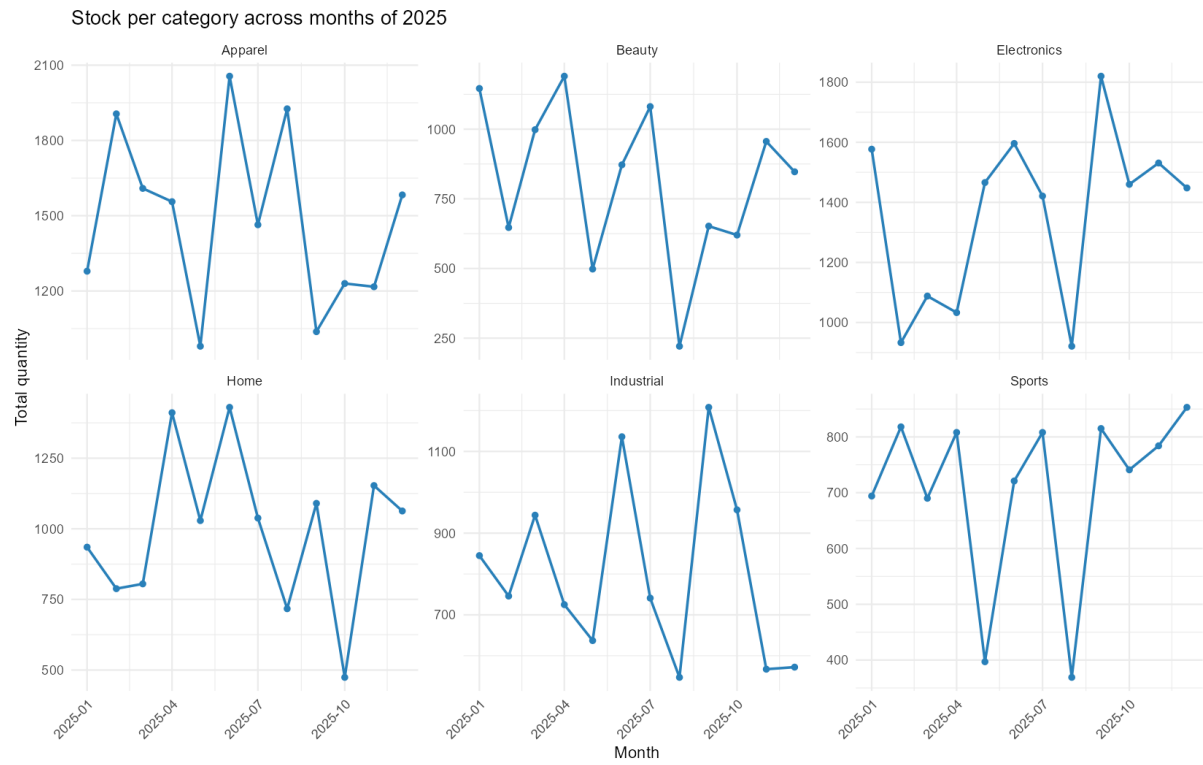


Figure 7: Stock per Category Across 2025 Months

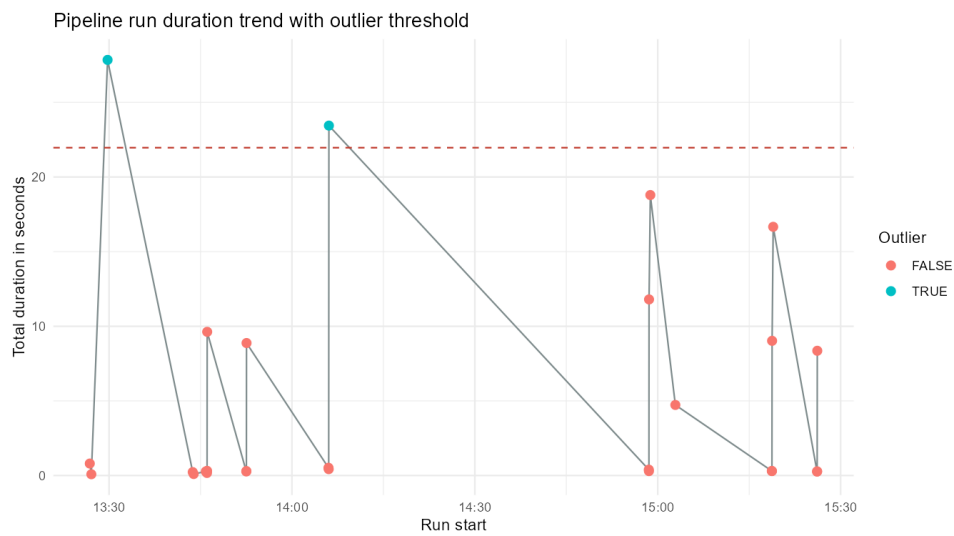


Figure 8: Run Duration Trend with Outlier Threshold

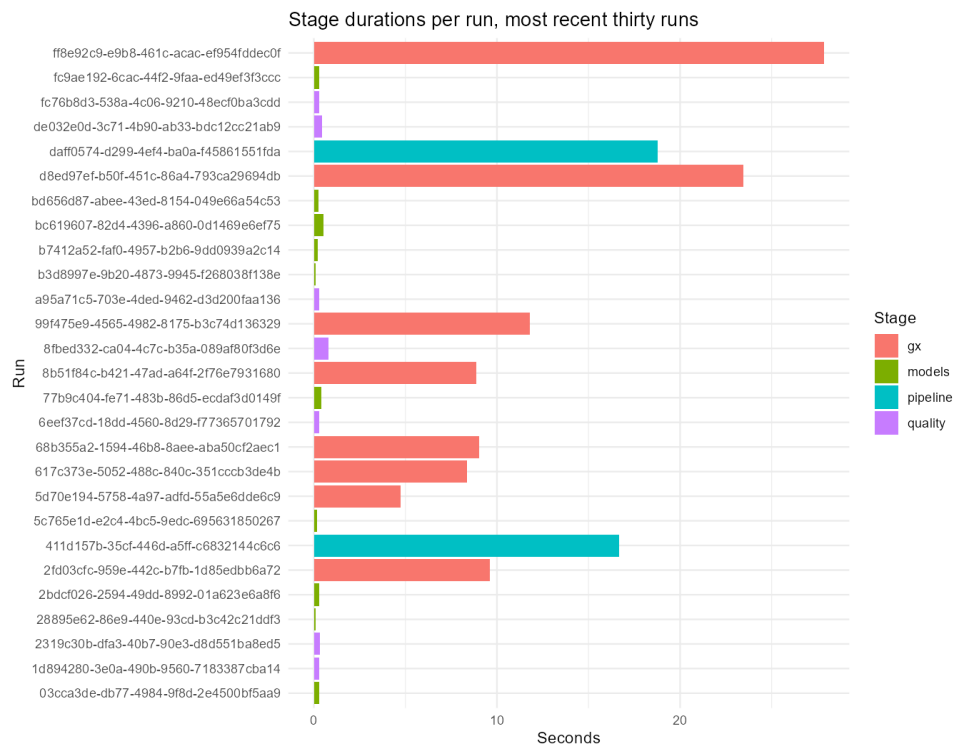


Figure 9: Stage Durations per Run

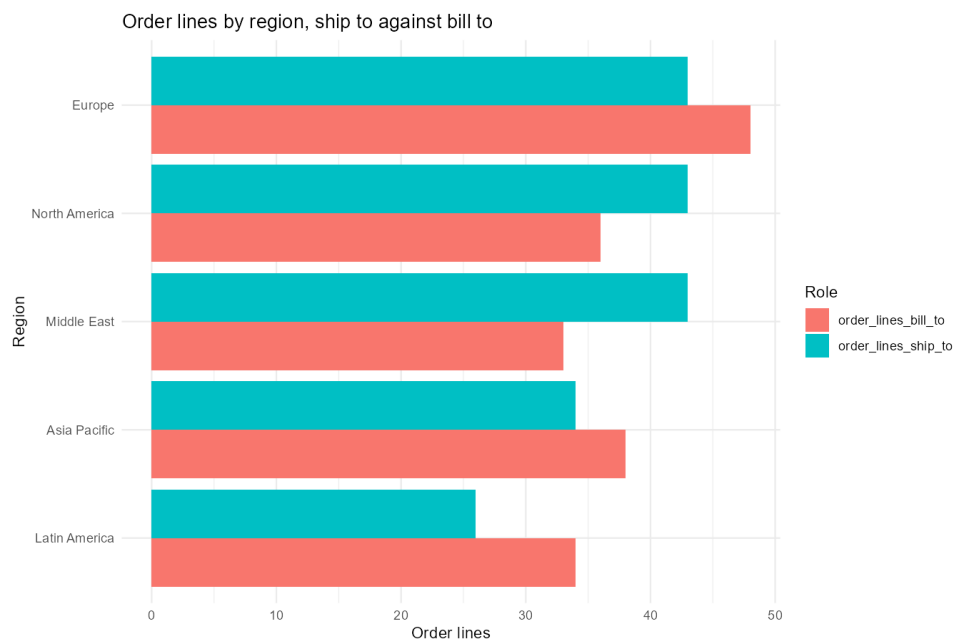


Figure 10: Order Lines by Region, Ship To Against Bill To